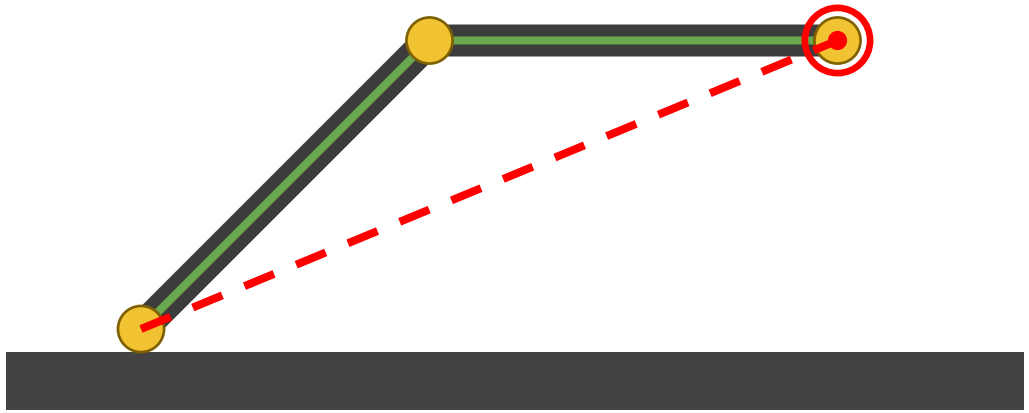
The Trigonometric Approach to Inverse Kinematics

Robotic Arm Course · Lesson 6

Day 1: The flat arm with triangles
Day 2: Adding the base turn (3D)

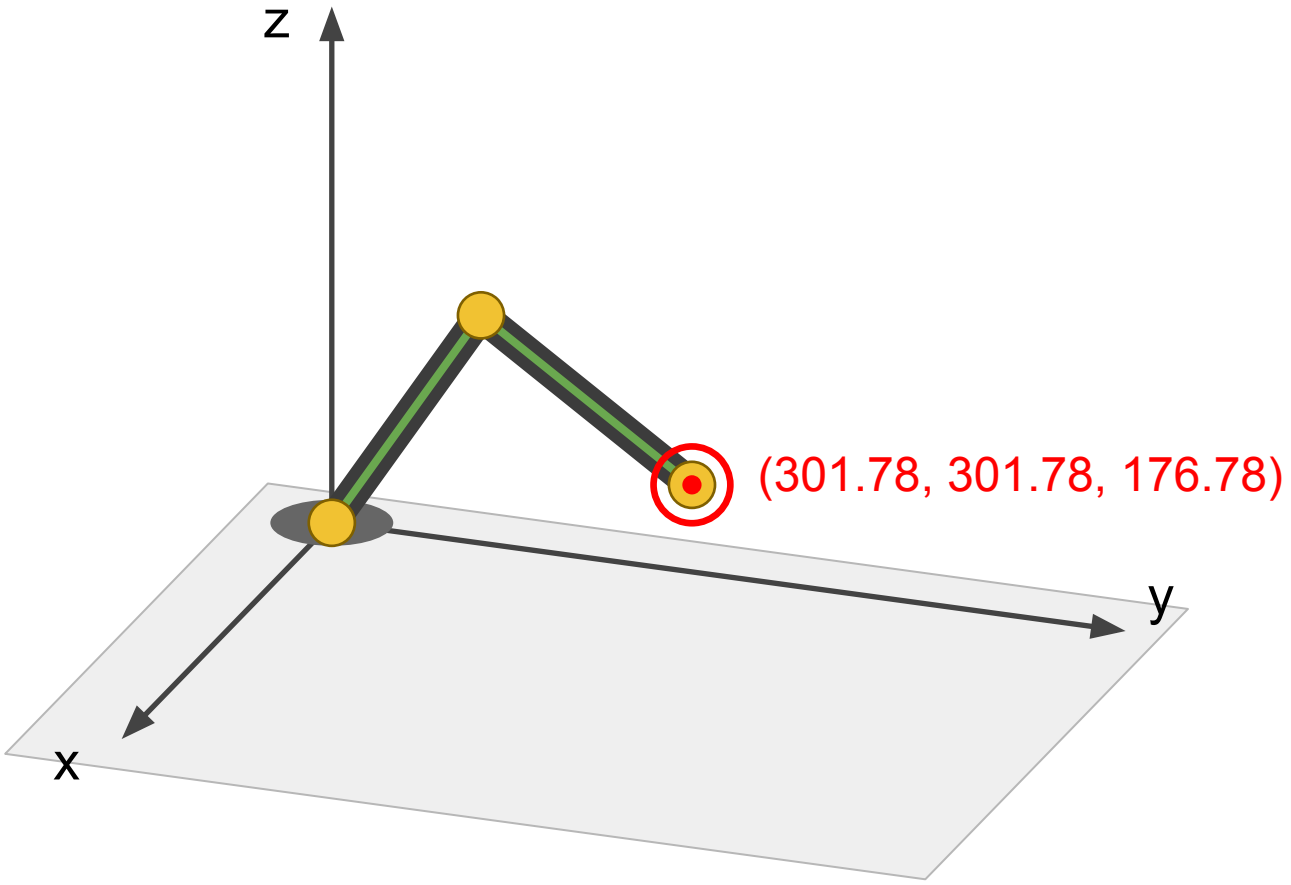


Where Lesson 5 left off:

Forward kinematics (FK):
angles in, tip position out.

tip_3d(45, 45, -45)
gives [301.78, 301.78, 176.78]

Today we go the other way.
We know where the tip should go.
We find the angles.



Forward vs. inverse kinematics:

Forward kinematics (FK)
angles → tip position
Always exactly one answer.
Lessons 3, 4 and 5.

Inverse kinematics (IK)
tip position → angles
There can be no answer, one or two.
Lessons 6, 7 and 8.

A robot is told a target, like “pick up the cup at (x, y, z)”. Its motors need angles, not positions. IK is the step that turns the target into angles.

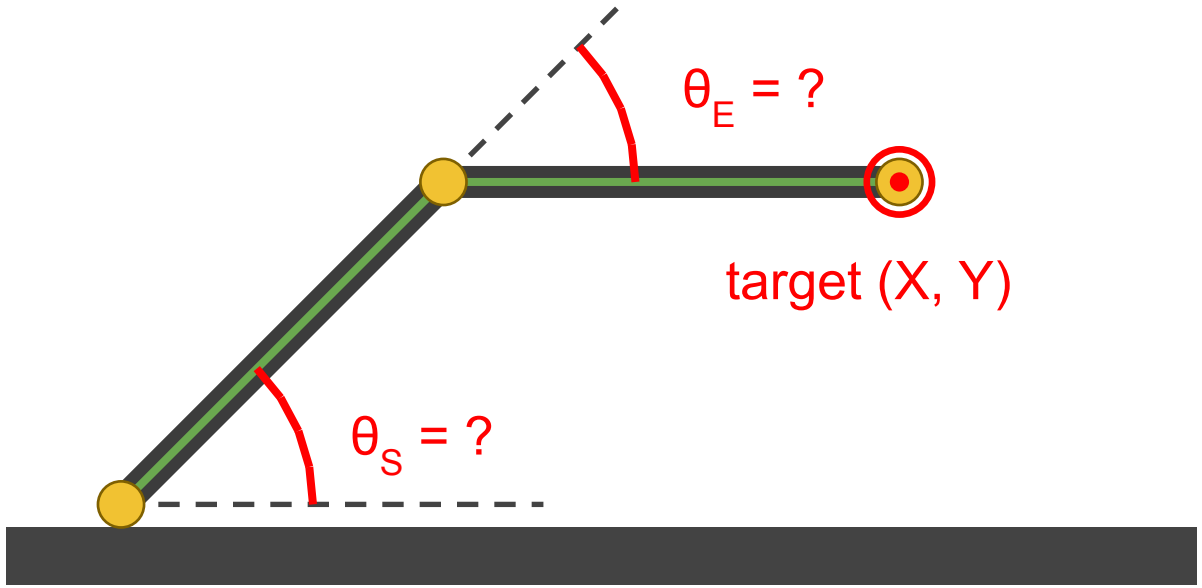
Lesson 3 used triangles to go forward. Today we use the same triangles backwards.

The IK problem (flat arm):

We know:
$L_1 = L_2 = 250$ mm
the target (X, Y)

We want:
θ_S and θ_E

We solve the flat arm first. On Day 2 we add the base turn.



The law of cosines:

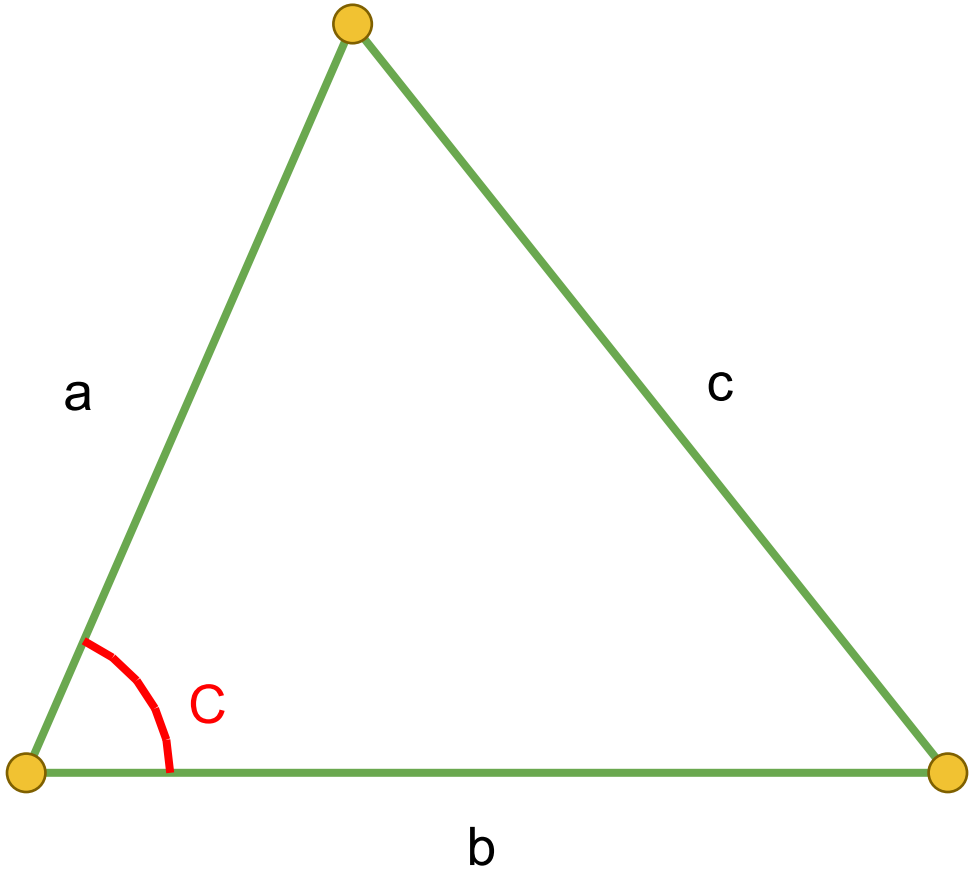
For any triangle with sides a, b and c, where C is the angle opposite side c:

$c^2 = a^2 + b^2 - 2ab \cos C$

Solve it for the angle:
$\cos C = (a^2 + b^2 - c^2) \div (2ab)$

Use it when you know all three sides and want an angle.

If C = 90°, then cos C = 0 and it becomes the Pythagorean theorem: $c^2 = a^2 + b^2$.



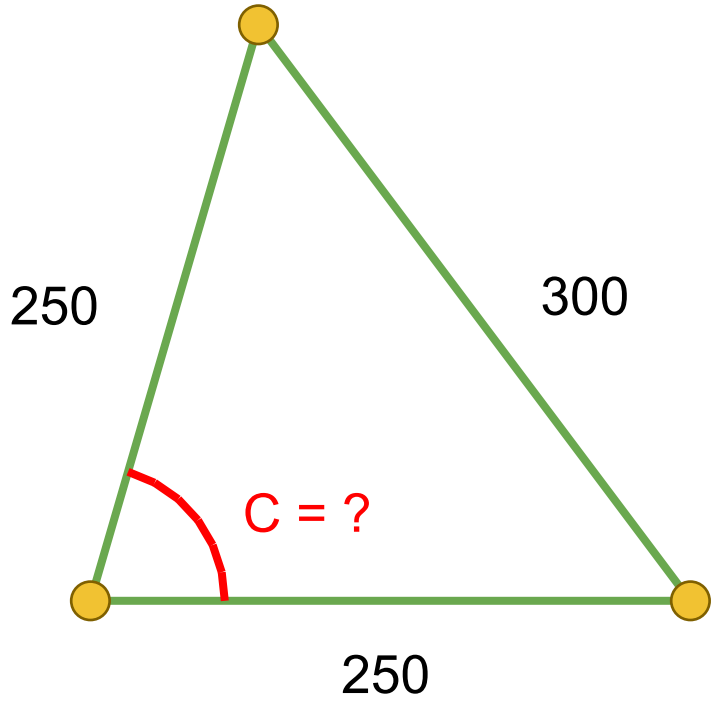
Worked example: the law of cosines

A triangle has a = 250, b = 250 and c = 300.
Find the angle C.

$\cos C = (250^2 + 250^2 - 300^2) \div (2 \times 250 \times 250)$
$= (125{,}000 - 90{,}000) \div 125{,}000$
$= 0.28$

$C = \cos^{-1}(0.28) = 73.74°$

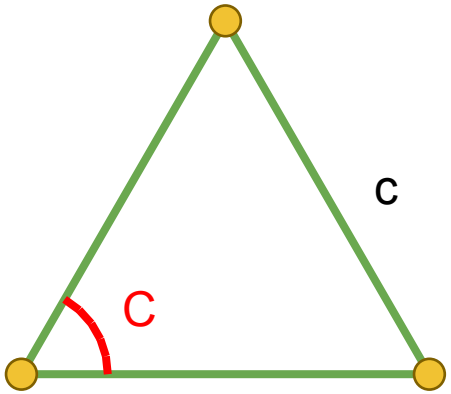
$\cos^{-1}$ is the inverse cosine key (also called arccos).
Set your calculator to degrees.



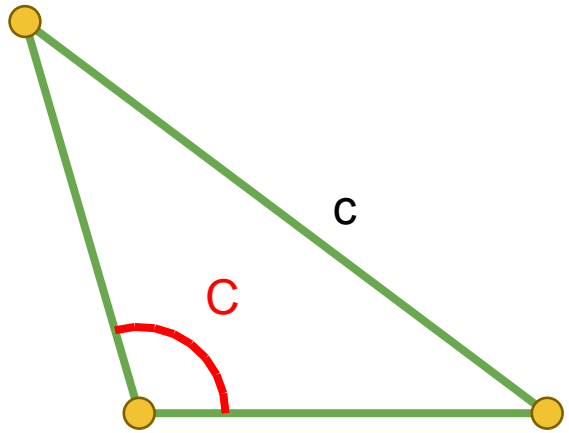
Try it yourself #1:

Find the angle C opposite side c. In both triangles a = b = 250.

(a) c = 250



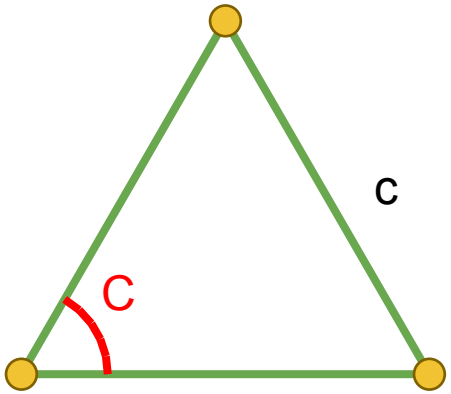
(b) c = 400



Try it yourself #1: solution

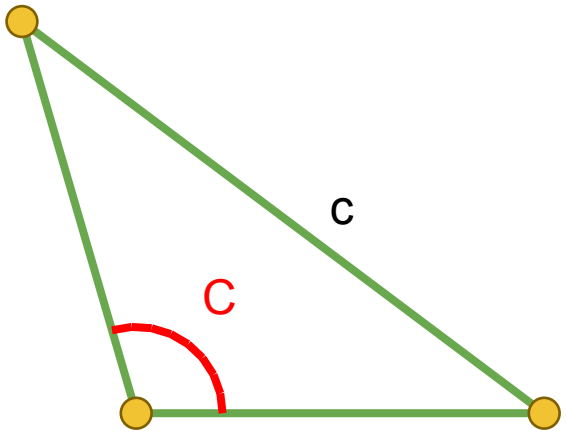
Find the angle C opposite side c. In both triangles a = b = 250.

(a) c = 250



cos C = (125,000 − 62,500) ÷ 125,000
= 0.5
C = $\cos^{-1}(0.5) = 60°$

(b) c = 400



cos C = (125,000 − 160,000) ÷ 125,000
= −0.28
C = $\cos^{-1}(-0.28) = 106.26°$

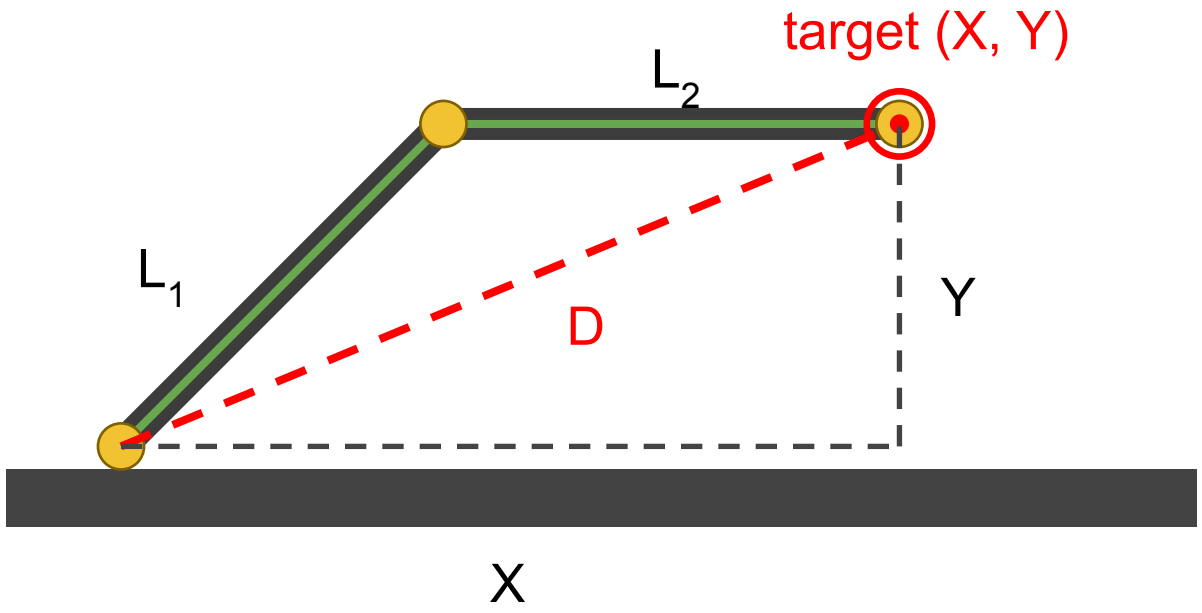
A negative cosine means the angle is bigger than 90°.

The arm makes a triangle:

Draw a straight line D from the shoulder to the target.

The two links and D make a triangle. We know all three sides:
$L_1 = 250$ and $L_2 = 250$
$D = \sqrt{(X^2 + Y^2)}$ (Pythagorean theorem)

So the law of cosines can find the angles of this triangle. They give us the joint angles.



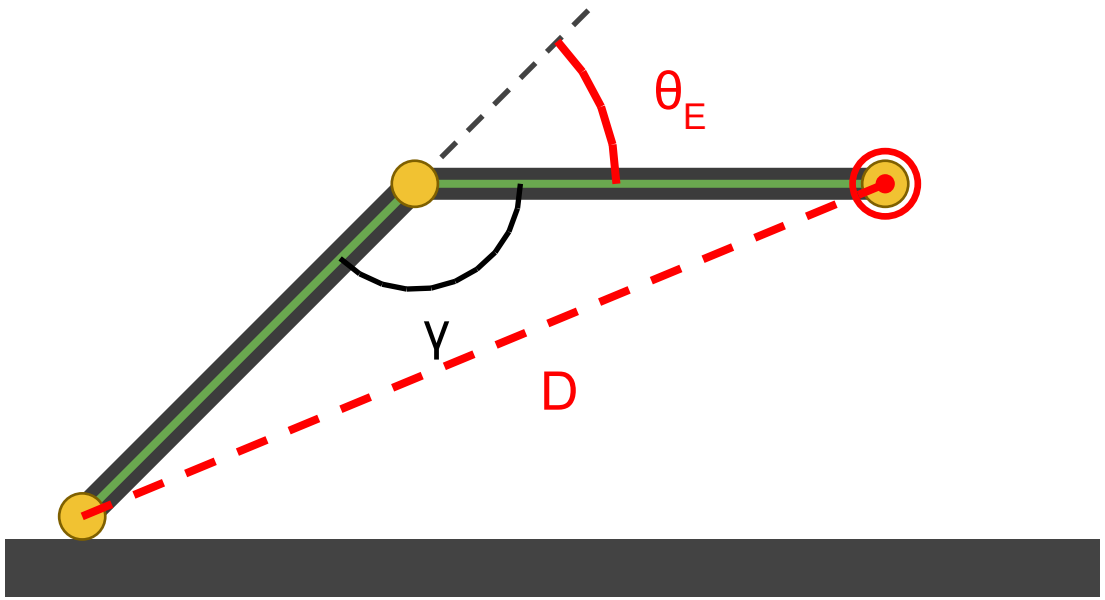
Step 1: the elbow angle θ_E

The law of cosines gives γ, the angle inside the triangle at the elbow:
$\cos \gamma = (L_1^2 + L_2^2 - D^2) \div (2\, L_1\, L_2)$

θ_E is the bend away from a straight arm, so its size is 180° − γ. Because cos(180° − γ) = −cos γ:
$\cos \theta_E = (D^2 - L_1^2 - L_2^2) \div (2\, L_1\, L_2)$

Check: a straight arm has D = 500, so $\cos \theta_E = 1$ and $\theta_E = 0°$.

Our forearm bends down (elbow up), so θ_E is negative.



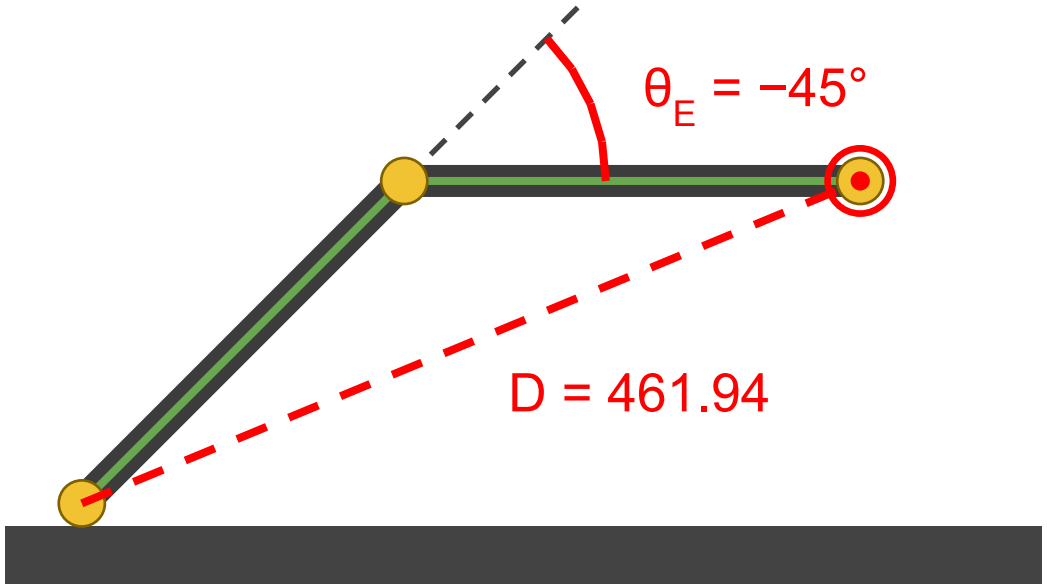
Worked example: the elbow

Target (X, Y) = (426.78, 176.78), the tip from Lesson 4.

$D^2 = 426.78^2 + 176.78^2 = 213{,}392$
(so $D = 461.94$)

$\cos \theta_E = (213{,}392 - 62{,}500 - 62{,}500) \div 125{,}000$
$= 0.7071$

$\cos^{-1}(0.7071) = 45°$. Elbow up, so $\theta_E = -45°$



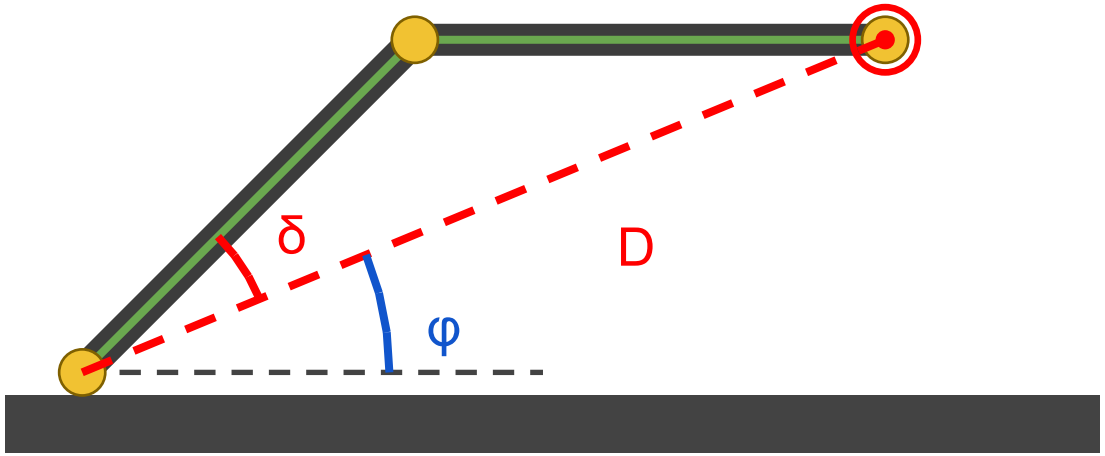
Step 2: the shoulder angle θ_S

θ_S is measured from the floor. It is made of two angles.

φ is the tilt of the line D:
$\varphi = \tan^{-1}(Y \div X)$

δ is the angle between D and the upper arm (the law of cosines again):
$\cos \delta = (L_1^2 + D^2 - L_2^2) \div (2\, L_1\, D)$
Our arm has $L_1 = L_2$, so this is just $\cos \delta = D \div 500$.

Elbow up: the upper arm is above D, so $\theta_S = \varphi + \delta$



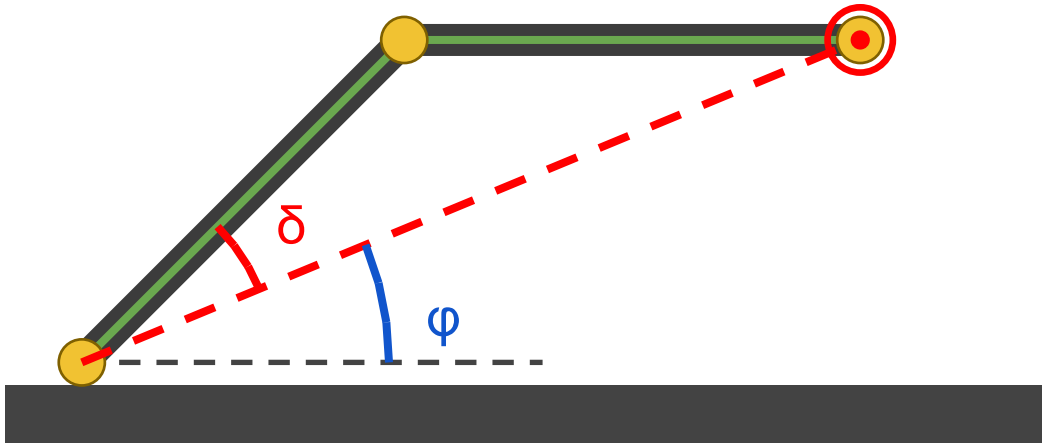
Worked example: the shoulder

Same target (426.78, 176.78), with D = 461.94.

$\varphi = \tan^{-1}(176.78 \div 426.78) = 22.5°$

$\cos \delta = (62{,}500 + 213{,}392 - 62{,}500) \div (2 \times 250 \times 461.94)$
$= 213{,}392 \div 230{,}970 = 0.9239$
(the same as $461.94 \div 500$)
$\delta = \cos^{-1}(0.9239) = 22.5°$

$\theta_S = 22.5° + 22.5° = 45°$



The answer:

IK found $\theta_S = 45°$ and $\theta_E = -45°$.

These are the angles Lesson 4 started with.

Check with FK (Lesson 5):

```
show(flat_tip(45, -45))
```

```
[426.78, 176.78]
```

FK lands back on the target, so the angles are right.

Step	Value
D	461.94
$\cos \theta_E$	0.7071
θ_E	−45°
φ	22.5°
δ	22.5°
θ_S	45°

Try it yourself #2:

Find θ_S and θ_E (elbow up) for the target (341.51, 341.51). $L_1 = L_2 = 250$ mm.

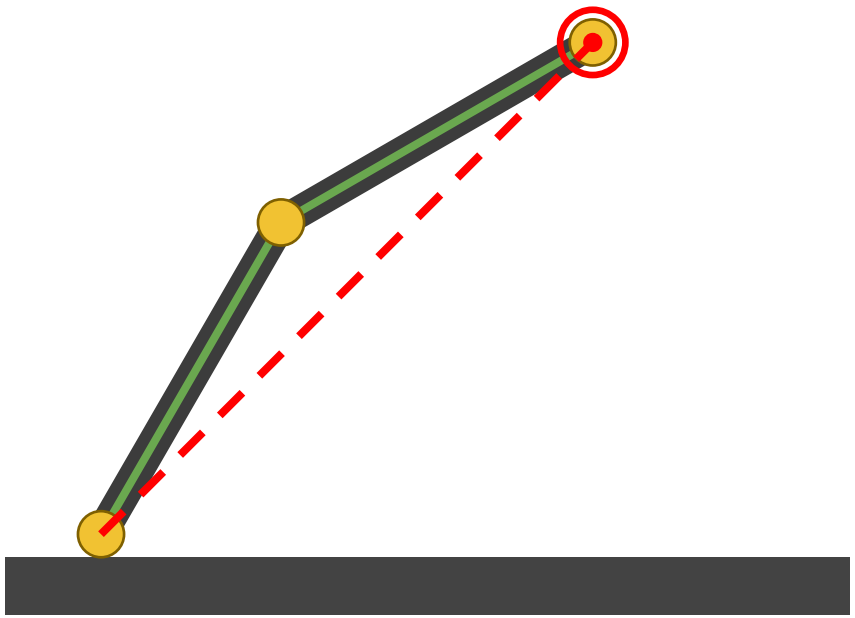
Try it yourself #2: solution

Find θ_S and θ_E (elbow up) for the target (341.51, 341.51). $L_1 = L_2 = 250$ mm.

$D^2 = 341.51^2 + 341.51^2 = 233{,}258$ $(D = 482.97)$

$\cos \theta_E = (233{,}258 - 125{,}000) \div 125{,}000 = 0.8661$
$\rightarrow \theta_E = -30°$

$\varphi = \tan^{-1}(341.51 \div 341.51) = 45°$
$\cos \delta = 482.97 \div 500 = 0.9659 \rightarrow \delta = 15°$
$\rightarrow \theta_S = 45° + 15° = 60°$



This is Lesson 4's Try it yourself #2 run backwards. Check: flat_tip(60, -30) gives [341.51, 341.51].

Try it yourself #3:

Find θ_S and θ_E (elbow up) for the target (250, 250). $L_1 = L_2 = 250$ mm.

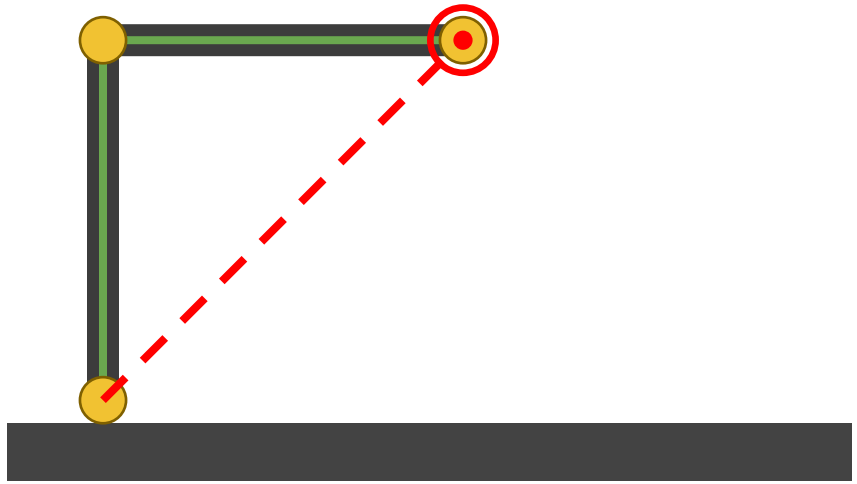
Try it yourself #3: solution

Find θ_S and θ_E (elbow up) for the target (250, 250). $L_1 = L_2 = 250$ mm.

$D^2 = 250^2 + 250^2 = 125{,}000$ $(D = 353.55)$

$\cos \theta_E = (125{,}000 - 125{,}000) \div 125{,}000 = 0$
$\rightarrow \theta_E = -90°$

$\varphi = \tan^{-1}(250 \div 250) = 45°$
$\cos \delta = 353.55 \div 500 = 0.7071 \rightarrow \delta = 45°$
$\rightarrow \theta_S = 45° + 45° = 90°$



The upper arm points straight up and the forearm points straight across. Check: flat_tip(90, -90) gives [250.0, 250.0].

Elbow up or elbow down:

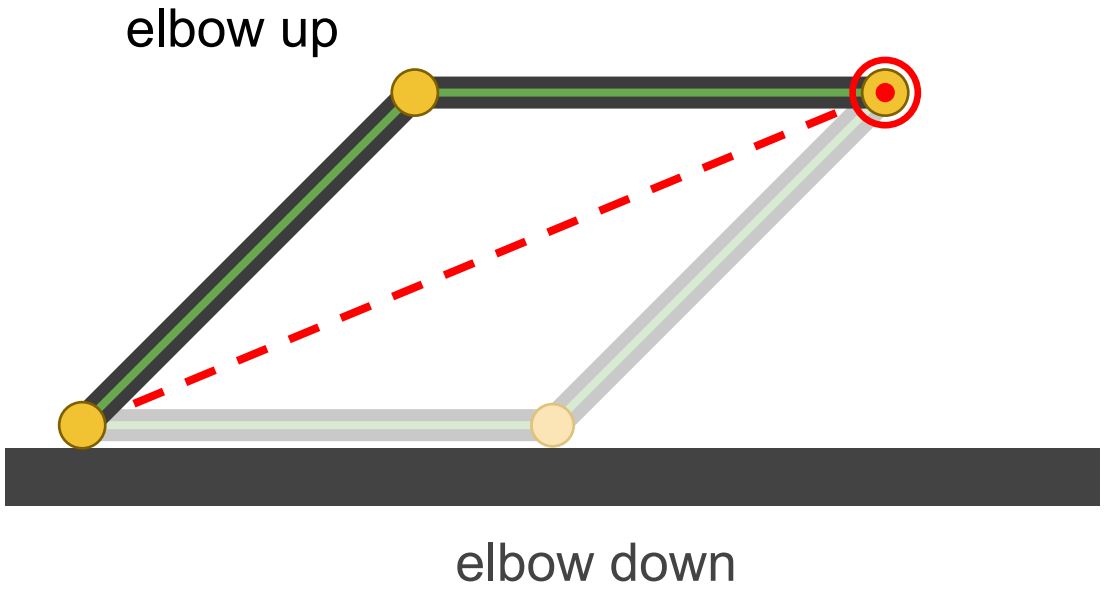
$\cos^{-1}$ only gives a positive angle (here 45°). We choose the sign of θ_E.

Elbow up: $\theta_E = -45°$ and $\theta_S = \varphi + \delta = 45°$

Elbow down: $\theta_E = +45°$ and $\theta_S = \varphi - \delta = 0°$

Both put the tip on (426.78, 176.78).

In this course we use elbow up. Check the real arm's joint limits before you use elbow down.

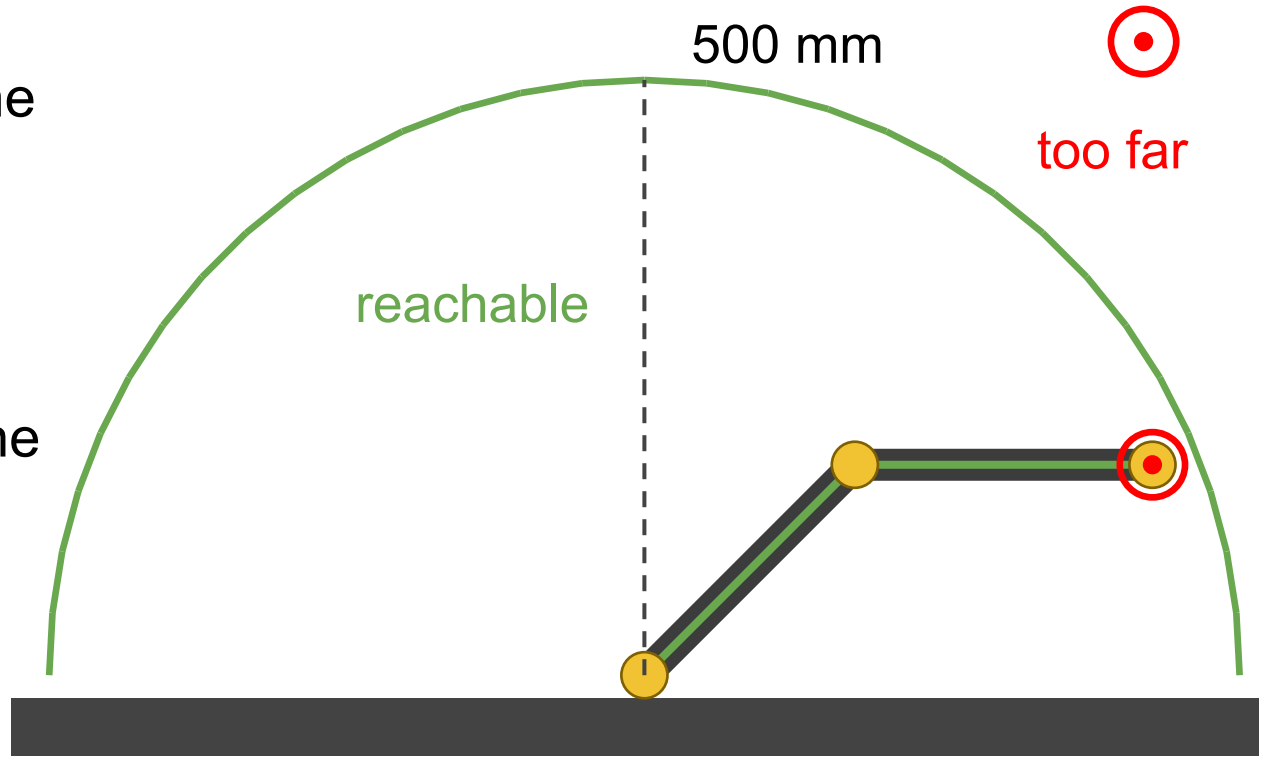


Can the arm reach it?

A triangle can only close if $D \leq L_1 + L_2 = 500$ mm.

If D is bigger, $\cos \theta_E$ comes out bigger than 1, and $\cos^{-1}$ gives an error. No angle has a cosine bigger than 1.

So always check $\cos \theta_E$ first:
between −1 and 1: the target is reachable.
anything else: the target is out of reach, and the robot must not move.



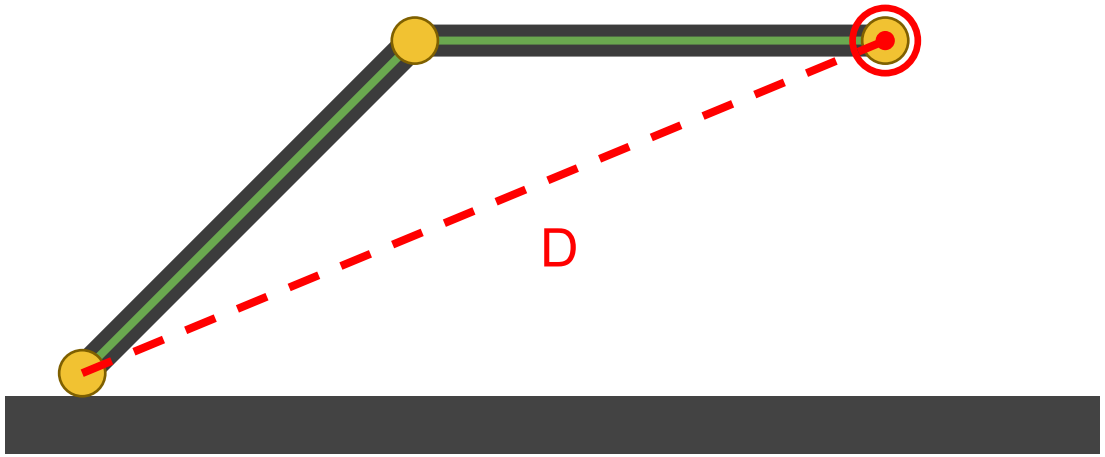
Day 1 recap:

1. Triangle: the two links and the line D, with $D = \sqrt{(X^2 + Y^2)}$.
2. Elbow: $\cos \theta_E = (D^2 - L_1^2 - L_2^2) \div (2\, L_1\, L_2)$. Elbow up, so θ_E is negative.
3. Shoulder: $\theta_S = \varphi + \delta$, with $\varphi = \tan^{-1}(Y \div X)$ and $\cos \delta = (L_1^2 + D^2 - L_2^2) \div (2\, L_1\, D)$.
4. Checks: $\cos \theta_E$ must be between −1 and 1. Then check the answer with FK.

Tomorrow the base turns too, so we need 3D.

Day 2: where Day 1 left off

1. What is θ_E for a straight arm?
2. What is the largest D the arm can reach?
3. What is the sign of θ_E for elbow up?



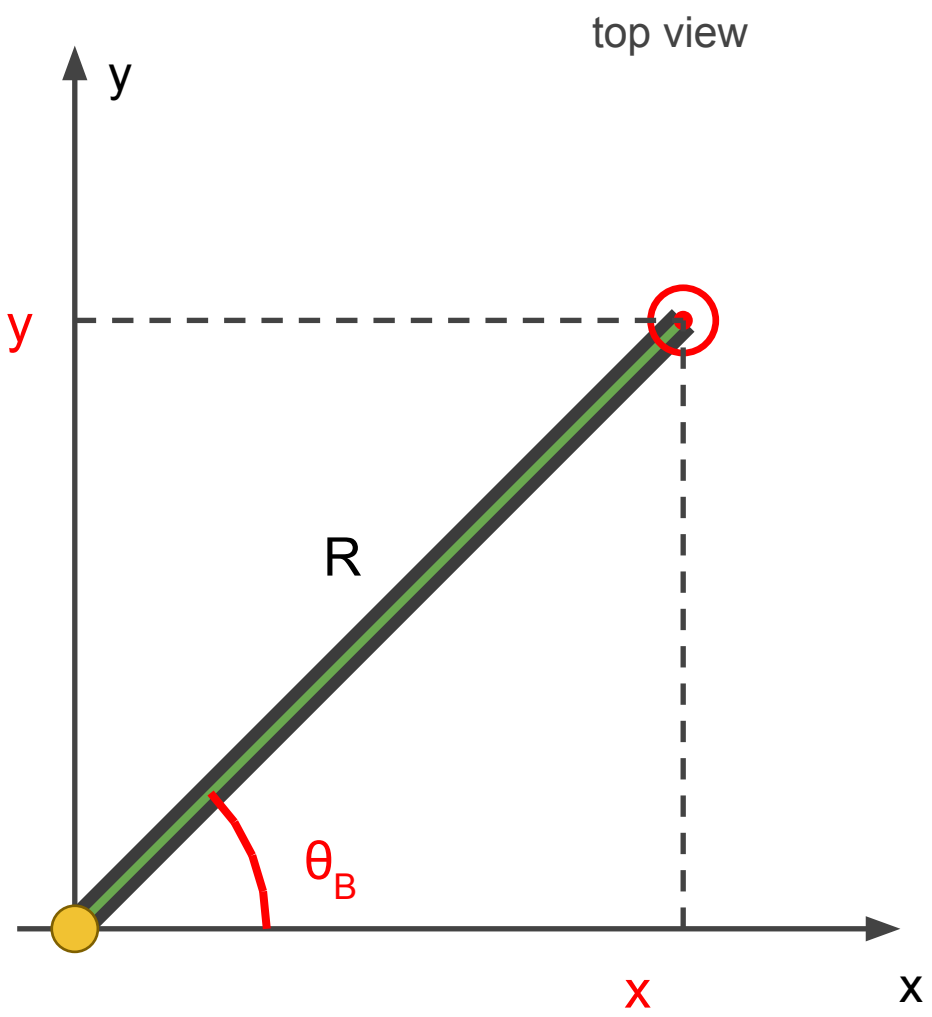
Step 1: the base aims the arm

Look from above. The base turns until the arm points at the target's (x, y) spot.

In Lesson 4: $x = R \cos \theta_B$ and $y = R \sin \theta_B$.
Divide: $y \div x = \tan \theta_B$, so
$\theta_B = \tan^{-1}(y \div x)$

Target (301.78, 301.78, 176.78):
$\theta_B = \tan^{-1}(301.78 \div 301.78) = \tan^{-1}(1) = 45°$

Careful: plain $\tan^{-1}$ can point the wrong way. See "Why atan2?".

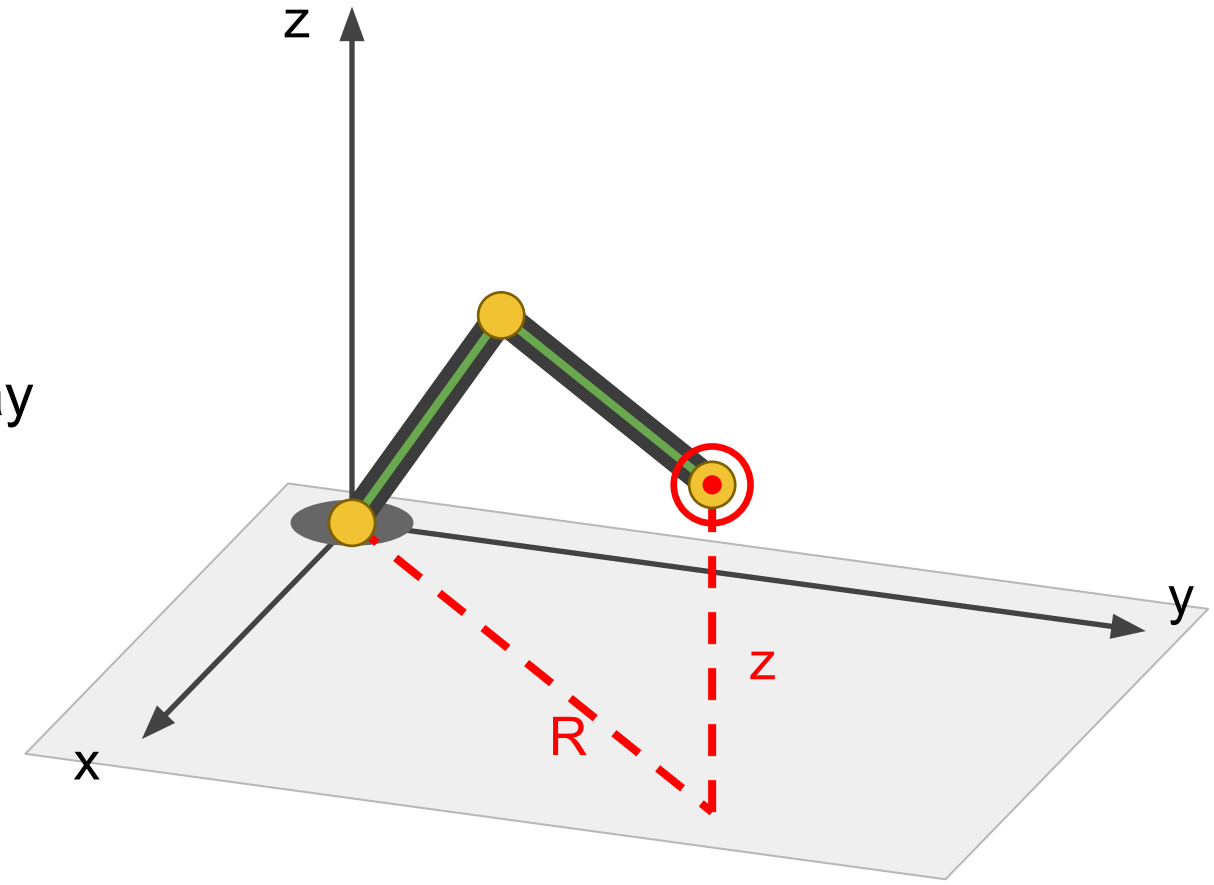


Step 2: flatten 3D into 2D

The arm always stays in one upright plane: the plane the base points at.

Measure the target inside that plane:
how far out: $R = \sqrt{(x^2 + y^2)}$
how high: z

Then (R, z) plays the role of (X, Y) from Day 1. The elbow and shoulder steps do not change.



Why atan2?

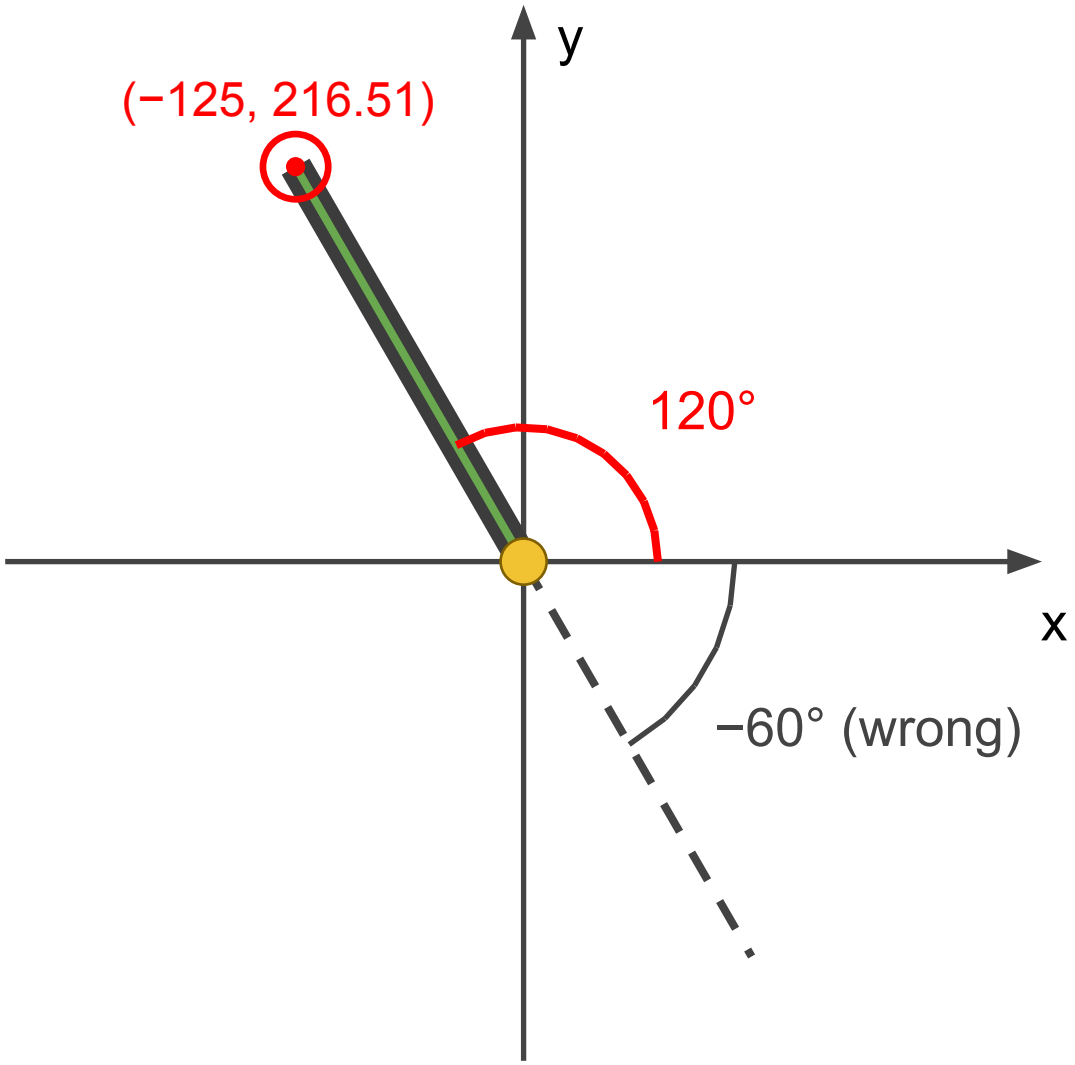
Try the spot (x, y) = (−125, 216.51).

$\tan^{-1}(216.51 \div -125) = \tan^{-1}(-1.732) = -60°$.
That points the wrong way!

y ÷ x is the same for (−125, 216.51) and (125, −216.51), so $\tan^{-1}$ cannot tell them apart.

atan2(y, x) looks at the signs of both y and x, so it picks the right one:
atan2(216.51, −125) = 120°

In Python: math.degrees(math.atan2(y, x))



The full recipe:

1. Aim the base: $\theta_B = \text{atan2}(y, x)$

2. Flatten: $R = \sqrt{(x^2 + y^2)}$, then $D^2 = R^2 + z^2$

3. Elbow: $\cos \theta_E = (D^2 - L_1^2 - L_2^2) \div (2\, L_1\, L_2)$
   Check that $\cos \theta_E$ is between −1 and 1. Elbow up, so θ_E is negative.

4. Shoulder: $\varphi = \tan^{-1}(z \div R)$ and $\cos \delta = (L_1^2 + D^2 - L_2^2) \div (2\, L_1\, D)$
   $\theta_S = \varphi + \delta$

Four steps, a few formulas, and an exact answer. No guessing.

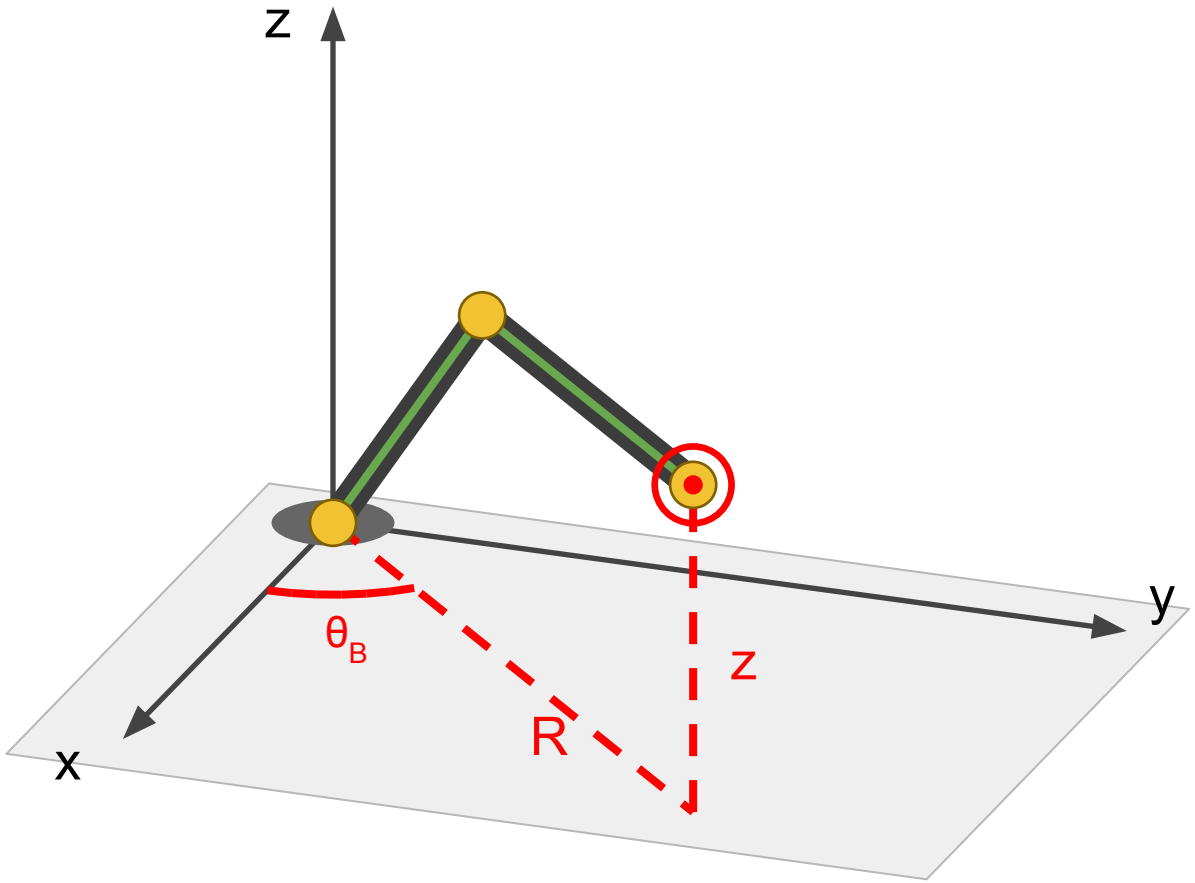
Worked example:

Find θ_B, θ_S and θ_E for the target (301.78, 301.78, 176.78), the tip from Lesson 4.

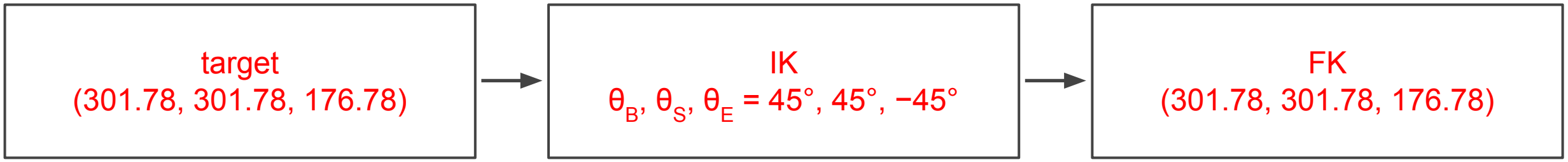
1. θ_B = atan2(301.78, 301.78) = 45°
2. R = $\sqrt{(301.78^2 + 301.78^2)}$ = 426.78
   z = 176.78

Now it is the flat arm with target (426.78, 176.78): Day 1's worked example!

3. θ_E = −45°
4. θ_S = 45°



Check it with FK:



Put the IK angles back into FK. If the tip lands on the target, the angles are right.

Three ways to do FK:

1. Lesson 4's link table, by hand.
2. The Desmos graph from Lesson 4: u = 45, v = 45, w = −45 gives d = (3.0178, 3.0178, 1.7678), so (301.78, 301.78, 176.78) mm.
3. Lesson 5's code: show(tip_3d(45, 45, -45)) prints [301.78, 301.78, 176.78].

Make it a habit: always check IK with FK.

Try it yourself #4:

Find θ_B, θ_S and θ_E (elbow up) for the target (295.75, 170.75, 91.51).

Use the four-step recipe. Hint: this tip came from Lesson 4's Try it yourself #5.

Try it yourself #4: solution

Find θ_B, θ_S and θ_E (elbow up) for the target (295.75, 170.75, 91.51).

1. $\theta_B = \text{atan2}(170.75, 295.75) = 30°$
2. $R = \sqrt{(295.75^2 + 170.75^2)} = 341.50$ and $z = 91.51$
   $D^2 = 341.50^2 + 91.51^2 \approx 125{,}000$ (D = 353.55)
3. $\cos \theta_E = (125{,}000 - 125{,}000) \div 125{,}000 = 0 \rightarrow \theta_E = -90°$
4. $\varphi = \tan^{-1}(91.51 \div 341.50) = 15°$, $\cos \delta = 353.55 \div 500 = 0.7071 \rightarrow \delta = 45°$
   $\rightarrow \theta_S = 15° + 45° = 60°$

Lesson 4's Try it yourself #5 run backwards: (30°, 60°, −90°). Check: show(tip_3d(30, 60, -90)) gives [295.75, 170.75, 91.51].

Try it yourself #5:

Find θ_B, θ_S and θ_E (elbow up) for the target $(-324.76, 187.50, 216.51)$.

Careful with θ_B: x is negative.

Try it yourself #5: solution

Find θ_B, θ_S and θ_E (elbow up) for the target (−324.76, 187.50, 216.51).

1. θ_B = atan2(187.50, −324.76) = 150° (plain $\tan^{-1}$ says −30°)
2. R = $\sqrt{(324.76^2 + 187.50^2)}$ = 375.00 and z = 216.51
   $D^2 = 375.00^2 + 216.51^2 = 187{,}502$ (D = 433.01)
3. $\cos \theta_E = (187{,}502 - 125{,}000) \div 125{,}000 = 0.5 \rightarrow \theta_E = -60°$
4. $\varphi = \tan^{-1}(216.51 \div 375.00) = 30°$, $\cos \delta = 433.01 \div 500 = 0.8660 \rightarrow \delta = 30°$
   $\rightarrow \theta_S = 30° + 30° = 60°$

Check: show(tip_3d(150, 60, -60)) gives [-324.76, 187.5, 216.51].

Try it yourself #6: can the arm reach it?

Find $D = \sqrt{(x^2 + y^2 + z^2)}$ for each target and compare it with 500 mm.

(a) (200, 200, 200) (b) (400, 300, 200)

Try it yourself #6: solution

Find $D = \sqrt{(x^2 + y^2 + z^2)}$ for each target and compare it with 500 mm.

(a) (200, 200, 200) (b) (400, 300, 200)

(a) $D = \sqrt{(40{,}000 + 40{,}000 + 40{,}000)} = 346.41$ mm. Less than 500: reachable.
($\cos \theta_E = -0.04$)

(b) $D = \sqrt{(160{,}000 + 90{,}000 + 40{,}000)} = 538.52$ mm. More than 500: out of reach.
$\cos \theta_E = 1.32$, which is impossible.

For (b) the robot should report an error instead of moving.

When IK gets tricky:

Case	Answers	Why
D > 500 mm	none	the triangle cannot close
D = 500 mm	one	the arm is fully straight, $\theta_E = 0°$
D < 500 mm	two	elbow up or elbow down
x = y = 0	infinitely many	the target is right above the base, so any θ_B works

FK always gives exactly one answer. IK can give none, one, two or infinitely many.

Near full reach, rounding matters: (433.01, 250.00) is really a straight arm, but the rounded numbers give $\theta_E = -0.35°$, not 0°. If rounding pushes the cosine just above 1 (like 1.001), treat it as 1. The arm is straight, not out of reach.

Trigonometry for IK: pros and cons

Pros

+ An exact answer from a few formulas
+ Fast: nothing repeats
+ Easy to check with FK

Cons

− New triangles for every new joint
− Special cases: quadrants, reach, two answers
− Only works if the arm can be drawn as triangles

If the arm had six joints, which method would you use for IK? Why?

Trigonometry is a great fit for our arm. Bigger arms need a method that does not depend on drawing triangles.

What we learned:

1. IK goes from a target to the joint angles. It can have no answer, one or two.

2. The arm makes a triangle, so the law of cosines gives θ_E and δ.

3. $\theta_B = \text{atan2}(y, x)$, and $R = \sqrt{(x^2 + y^2)}$ flattens 3D into 2D.

4. $\theta_S = \varphi + \delta$ (elbow up). Always check the answer with FK.

Next time (Lesson 7): gradient descent, a method that works for any arm. It guesses, measures the error and improves.